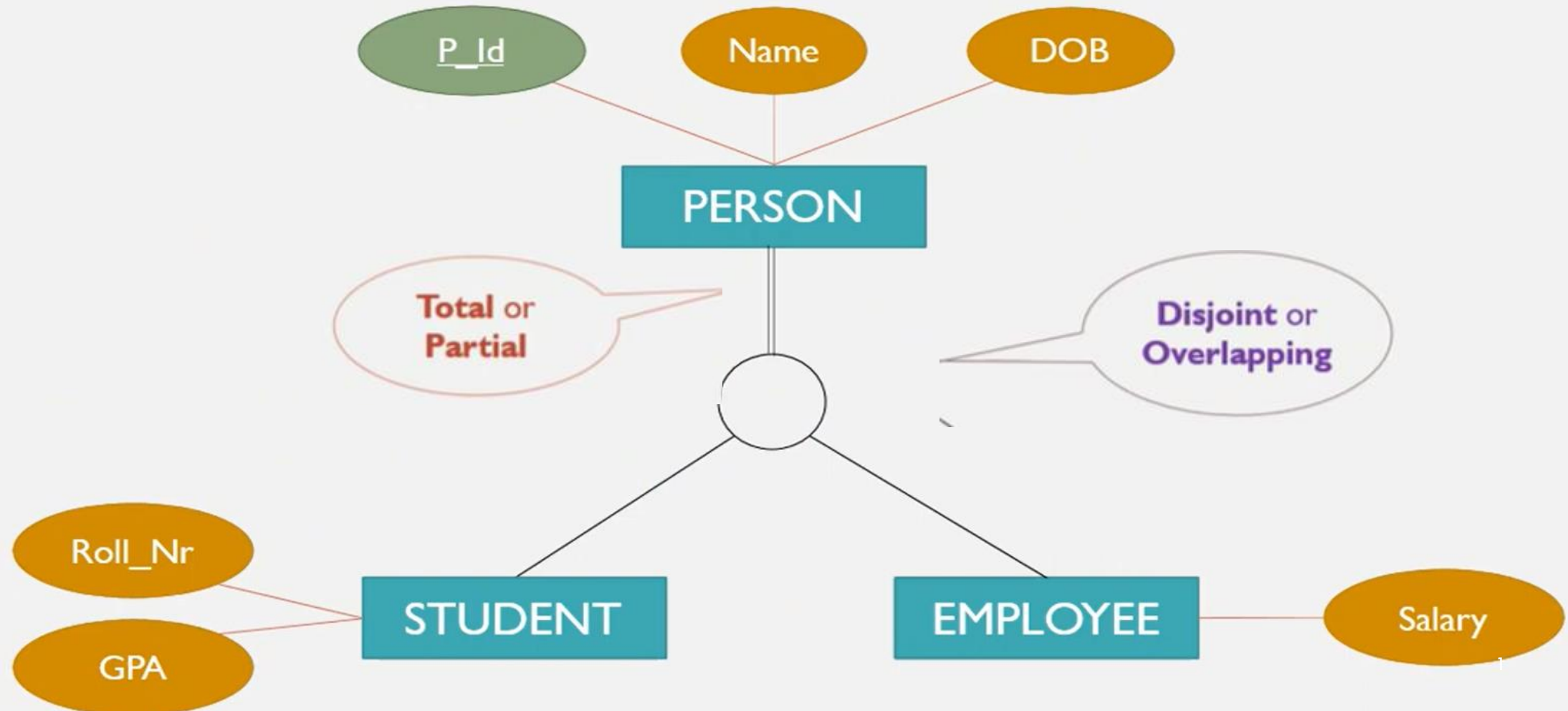


# SPECIALIZATION / GENERALIZATION



# STEP-8: CONVERT GENERALIZATION AND SPECIALIZATION

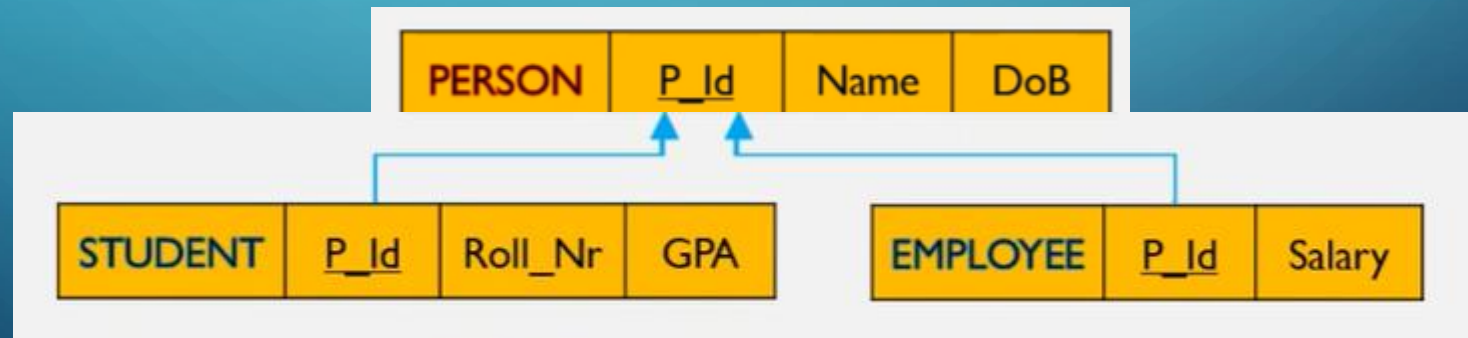
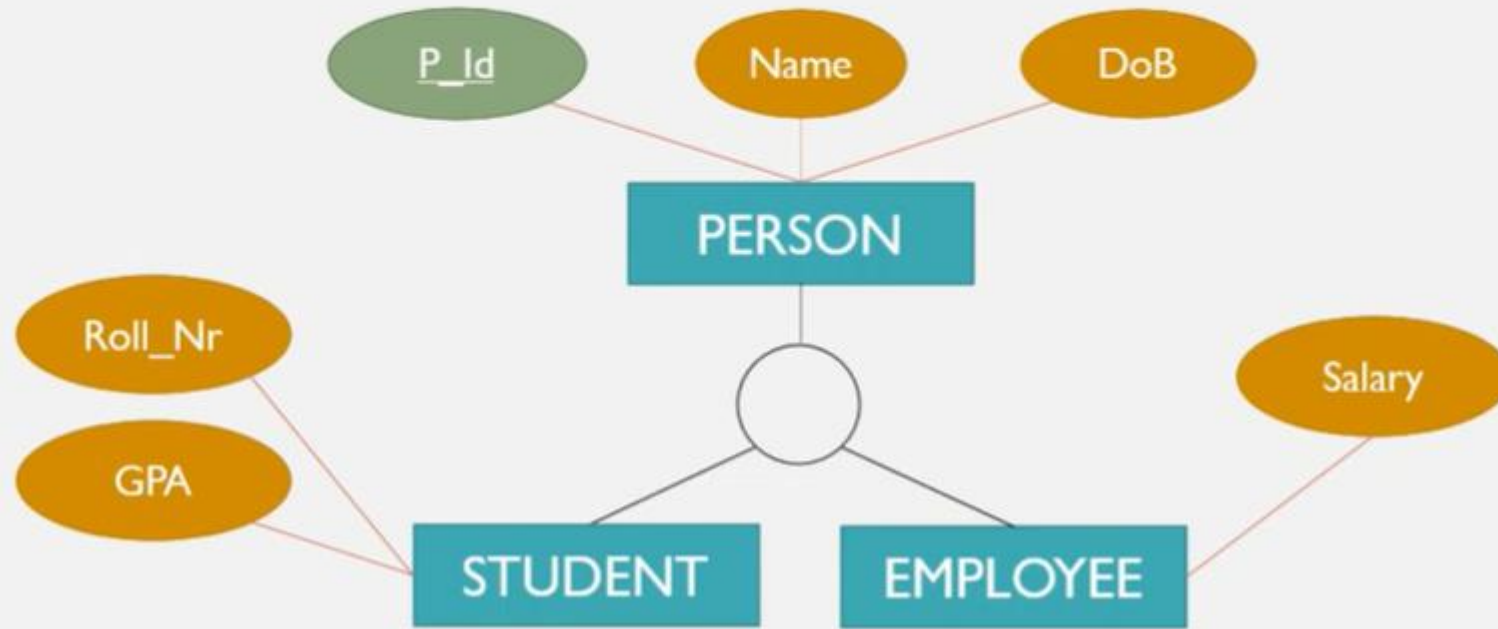
- A. Multiple Relations** – Both Superclass and Subclasses
- B. Multiple Relations** – Subclasses only
- C. Single Relation** – With one Type attribute
- D. Single Relation** – With multiple Type attributes

## STEP-8: OPTION → 8A

### [SUPERCLASS AND SUBCLASS]

- a) Create a relation for the **superclass** with all of its attributes.
- b) **Identifier/Key** of superclass will become the **primary key** of its relation.
- c) Create a separate relation for each **subclass** with respective attributes and include **primary key** from superclass relation as a **foreign key** in each relation.
  - a) This **foreign key** will become the **primary key** of each subclass relation.

## STEP-8: OPTION → 8A

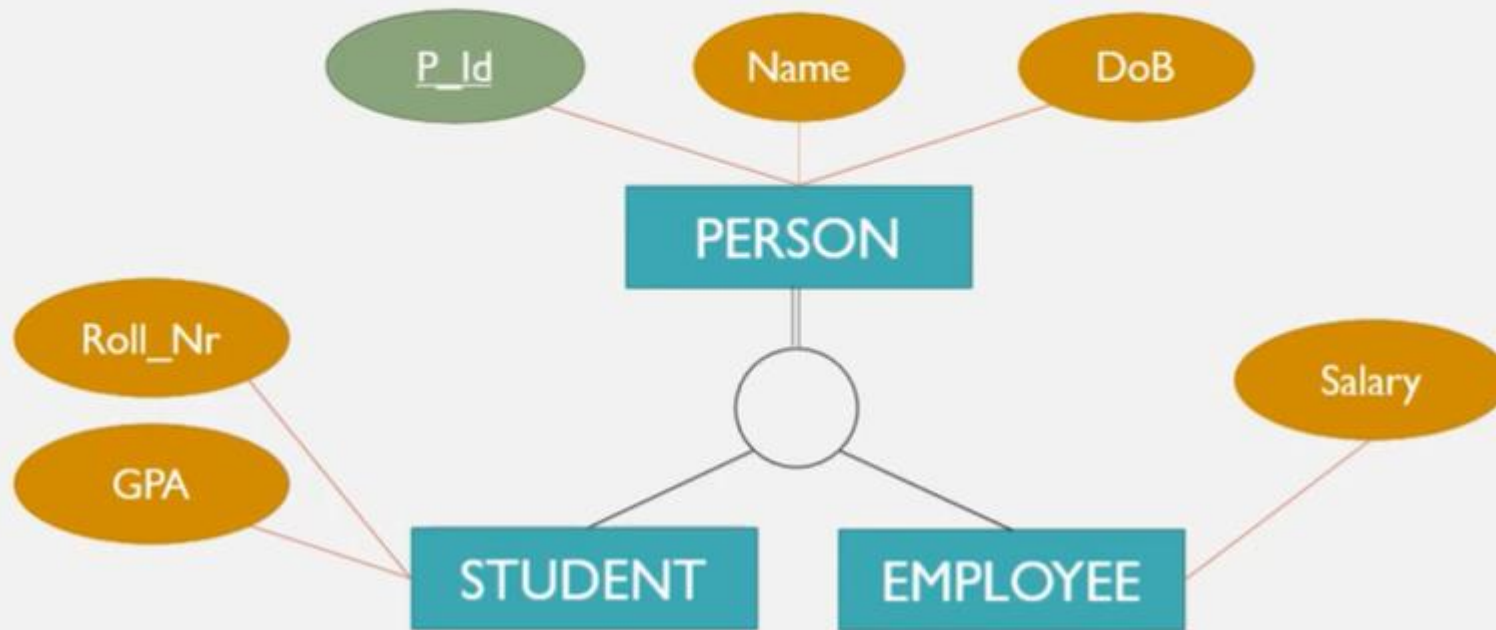




## STEP-8: OPTION → 8B [SUBCLASSES ONLY]

- a) Create a separate relation for each **subclass** with respective attributes and include all the attributes from the superclass.
- b) The **primary key** of superclass will become the **primary key** of each subclass relation.
- c) *This option only works for a specialization whose subclasses are **total**.*

## STEP-8: OPTION → 8B



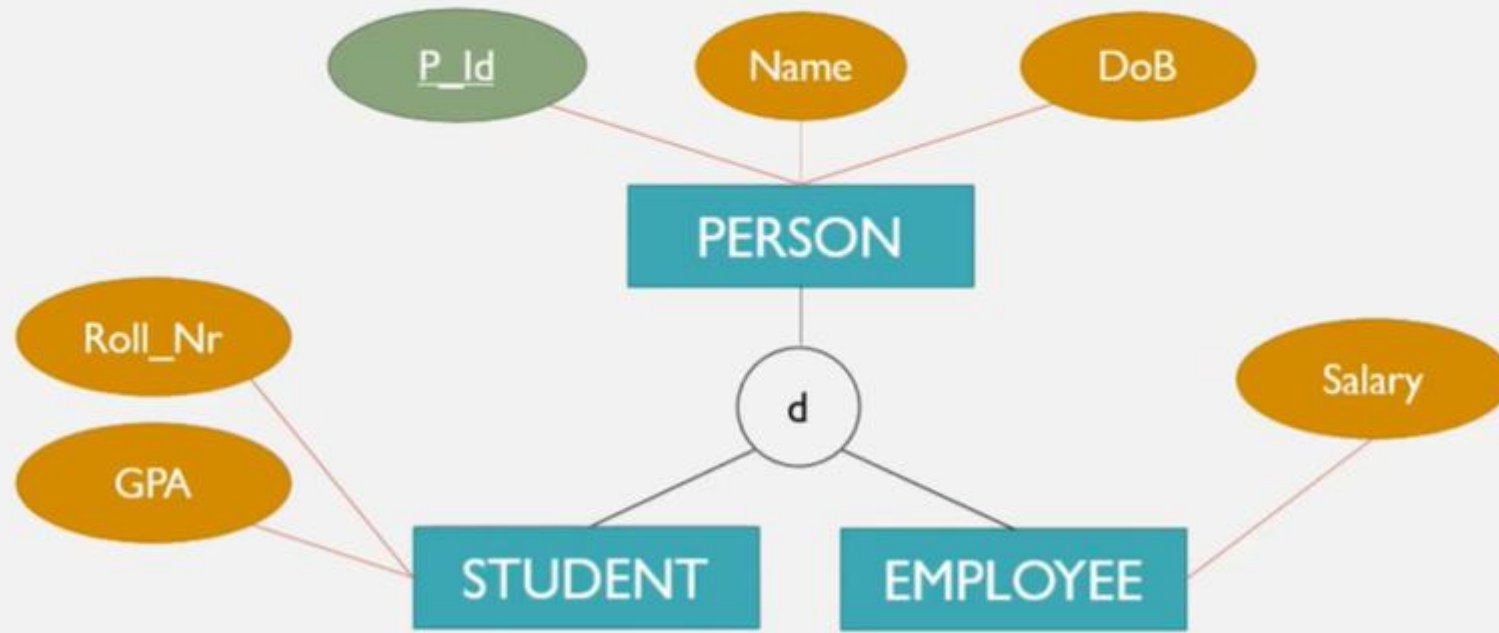
|         |             |      |     |         |     |
|---------|-------------|------|-----|---------|-----|
| STUDENT | <u>P_Id</u> | Name | DoB | Roll_Nr | GPA |
|---------|-------------|------|-----|---------|-----|

|          |             |      |     |        |
|----------|-------------|------|-----|--------|
| EMPLOYEE | <u>P_Id</u> | Name | DoB | Salary |
|----------|-------------|------|-----|--------|

## STEP-8: OPTION → 8C (TYPE ATTRIBUTE)

- a) Create a single relation that includes all the attributes from the **superclass** and each **subclass**.
- b) Add a new attribute **Type** to the relation.
  - a) It will indicate the subclass to which each tuple belongs.
- c) The **primary key** of superclass will become the **primary key** of relation.
- d) *This option only works for a specialization whose subclasses are **disjoint**.*

## STEP-8: OPTION → 8C



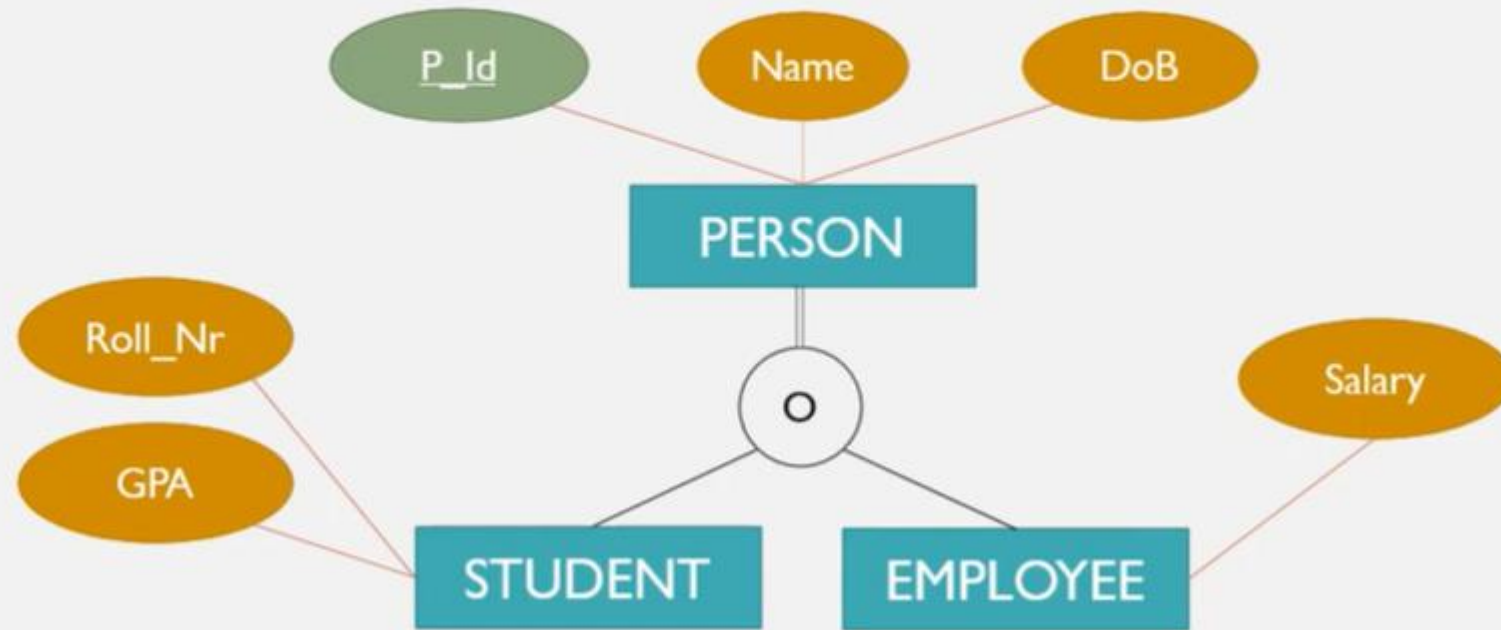
| PERSON | <u>P_Id</u> | Name | DoB | Type | Roll_Nr | GPA | Salary |
|--------|-------------|------|-----|------|---------|-----|--------|
|--------|-------------|------|-----|------|---------|-----|--------|



## STEP-8: OPTION → 8D [MULTIPLE TYPE ATTRIBUTES]

- a) Create a single relation that includes all the attributes from the **superclass** and each **subclass**.
- b) Add multiple **boolean Type** attribute to the relation (*one per each subclass*).
  - a) Each attribute indicates whether a tuple belongs to that subclass.
- c) The **primary key** of superclass will become the **primary key** of relation.
- d) *This option only works for a specialization whose subclasses are **overlapping**.*

## STEP-8: OPTION → 8D



| PERSON | <u>P_Id</u> | Name | DoB | FlagS | Roll_Nr | GPA | FlagE | Salary |
|--------|-------------|------|-----|-------|---------|-----|-------|--------|